Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12EE73]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER–2015

HIGH VOLTAGE ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions.
2. Missing data may be suitably assumed.

1.	**a.**	Mention the important applications of high voltage.	**06**
	b.	Explain the need of high voltage in the laboratory.	**04**
	c.	What is the usual classification of voltages used in a.c transmission?	**04**
2.	**a.**	Explain the working principles of a cascade transformer with a neat sketch.	**04**
	b.	Explain the construction and working of a series resonant circuit.	**04**
	c.	A 10 stage cockraft Walton circuit has all capacitors of 0.06 uf, the secondary voltage of the supply transformer is 100kv at a frequency of 150Hz. If the load current is 1MA, determine **i.** Voltage drop; **ii.** Ripple voltage **iii.** Optimum number of stages	**06**
3.	**a.**	Explain how impulse voltages are generated in laboratory using MARX circuit.	**07**
	b.	A 12 stage impulse generator has 0.126 resistances connected are 800Ω and 5000Ω respectively. If the load condenser is 1000PF, find the front and tail of the impulse wave produced.	**07**
4.	**a.**	With a neat sketch, explain the construction and working principle of electrostatic voltmeter. Bring out the advantages and disadvantages.	**07**
	b.	Discuss in detail the factors affecting the measurements of high voltage using sphere gap.	**07**
5.	**a.**	Discuss the method of straight detection for locating partial discharges in electrical equipment.	**07**
	b.	Write a short notes on : **i.** Rogowski- coil for high impulse current measurements **ii.** Magnetic link	**07**
6.	**a.**	What are the characteristics of switching surges?	**04**
	b.	What are effects of power frequency over voltages in power systems?	**06**

SCHOOL OF ENGINEERING & TECHNOLOGY
LIBRARY
DATE
JAIN UNIVERSITY-BANGALORE

	c.	How to control of over voltages due to switching?	**04**
7.	**a.**	State and explain Paschen's law with necessary diagram.	**06**
	b.	Write short notes on liquid dielectrics.	**08**
		i. Cavitation and Bubble theory	
		ii. Suspended particle theory	
8.		Explain the following tests:	
		i. High voltage tests on transformers	**04**
		ii. High voltage testing of cables	**04**
		iii. Testing of circuit breakers.	**06**